Instrument Floors

Three criteria for what a finite measurement determines,
with instruments and results on lattice $SU(2)$ and the zeros of ζ

Agent record, 21 October 2026. Byline unset — see **Contributions**.
Prior instrument: B. Wallace (TensorRent), FF06e.

Abstract

A finite battery of statistics applied to an object answers fewer questions than it appears to. We give three criteria for separating what a measurement determines from what its instrument contributed, state each as a proposition, build the instrument that tests it, and measure the result on two objects sharing no subject matter. **(I) Descent.** A surrogate construction is a choice of quotient $\pi\colon X \to X/G$; a statistic either factors through π or sees the fibre. Factoring statistics have a degenerate null distribution, so their standardised significance is $0/0$ rather than small; bounded statistics have a reachable ceiling on significance, below which no object can be convicted; and non-factoring is necessary but not sufficient for detection. A four-tier verdict follows. Applied to a 6×6 Wilson $SU(2)$ configuration, a battery of six statistics is shown to carry at most three independent readings. **(II) Linear invisibility in parameter space.** For a lattice action built from three $SU(2)$ characters, the entire quadratic fluctuation spectrum about the flat vacuum depends on one combination $c_1 = \beta - 2\alpha + 5\lambda$ of the three couplings; higher terms first enter at *quartic* order in the field. Two actions sharing c_1 are therefore indistinguishable to every linear measurement while differing in vacuum count — measured, with the discrepancy scaling as amplitude^4 over three halvings. **(III) Localisation.** A residual can be attributed to instrument truncation rather than to missing structure by masking the neighbourhoods where truncation rings and re-measuring. On the counting fluctuation of the Riemann zeros, explained variance rises from 0.790 to 0.959 on excluding points within a fifth of a mean spacing of a zero, while the excluded points fall to 0.400: the residual is instrument. The same mechanism accounts for an apparent loss of scale-stability, which becomes a decline of 0.003 over a fourfold range in height once the neighbourhoods are removed. Along the way the cubic character produces a first-order transition in the lattice with a measured hysteresis loop and two-phase coexistence at the predicted root, and we localise a known 0.4–1.0σ artifact in a published surrogate pipeline to piecewise-linear interpolation specifically, which makes it removable. All instruments are anchored on values they were not tuned to, and every figure is emitted from the measurement that produced it.

Status in this repository. A supporting document under `docs/`, outside the 35-paper canonical publication inventory and its build job, following the placement of the Yang–Mills comparison methodology paper. Its propositions and measurements are not part of the corpus author statement; see **Contributions** on the final page. Figures are generated by the script named in §13 into `instrument_floors_data.tex`, which this document `\inputs`.

1 Claim boundaries

Status is separated from exposition. No claim is stated more strongly in the text than here.

Claim	Status	Where
Props. 1–3, 4, 5	elementary, proved	§3, §5
Six lattice statistics carry ≤ 3 readings	measured; 4 of 4 collapses structural	§7
Equal c_1 hides a different vacuum count	measured; scaling exponent 4 confirmed	§5
First-order transition at $\rho = 1 + 5\lambda/\beta$	measured: loop, coexistence, spinodal	§6
The ζ residual is truncation, not structure	measured, one masking family	§8
Scale-stability holds after masking	measured, four windows	§8
The 0.4–1.0σ artifact needs linear interpolation	measured, two implementations	§10
Amplification: our point near the published curve	measured; converged 7% low, inside the fit band	§9
Any claim about ζ, RH, or continuum $SU(2)$	**none made**	§12

2 Introduction

An object admits no shortage of statistics, and when many of them behave as expected the usual reading is over-determination. The opposite reading is available on the same evidence: if the statistics are not independent, their agreement is redundancy, and if what they share is generic to a class rather than specific to the object, their convergence reports only class membership — true of uncountably many objects with nothing else in common.

That is one instance of a general problem. A measurement's output is a composite of the object and the instrument, and the instrument's contribution is itself a quantity to be measured rather than assumed small. Three cases arise repeatedly and admit sharp criteria:

1. A statistic may be *determined by a coarsening* of the object, so that it cannot distinguish the object from anything else with the same coarse data (§3).
2. A *parameter* of a model may be invisible to the order of response one is measuring, so that distinct models are indistinguishable by that measurement and differ elsewhere (§5).
3. A *residual* may be produced by the truncation of the instrument rather than by structure the model fails to reach, and the two are separable by where the residual lives (§8).

Each section below states the criterion, builds the instrument, anchors it, and reports what it measured. The two objects are a finite Wilson $SU(2)$ lattice gauge configuration and the first 2003 non-trivial zeros of ζ, chosen because they share no subject matter: a criterion validated on one object is accommodated to it.

Criterion I is an amendment to an existing instrument [2], itself a specialisation of surrogate-data testing [1]; criteria II and III are built here. None of the propositions is deep. The contribution is that each is omitted in practice, together with the measured size of the omission.

3 Criterion I: descent

3.1 Framework

Definition 1 (Quotient, surrogate)**.** *Let the object X decompose into components $x_1, \ldots, x_n$ and let $G = S_n$ act by permuting them. Write $\pi\colon X \mapsto X/G$ for the orbit map, which sends an object to its multiset of components — equivalently, to its one-point distribution, or* marginal*. A* surrogate *is gX for g drawn uniformly from G: identical marginal, no higher structure of X's.*

Definition 2 (Witness, significance, factoring)**.** *A* witness *is* $W\colon X \mapsto \mathbb{R}$*. Over an ensemble of* K *surrogates with mean* $\overline{W}$ *and standard deviation* σ_W*, the* standardised significance *is* $z(W) = |W(X) - \overline{W}|/\sigma_W$*.* W factors through the marginal *if* $W = \widetilde{W} \circ \pi$ *for some* $\widetilde{W}$*, equivalently if* $W(gX) = W(X)$ *for all* $g \in G$*.*

The existing criterion [2] declares W *form* when z is small ($\lesssim 3$) and *function* when large, reading the first as evidence that W responds to the class and the second that it responds to the object. The quotient language makes the content plain: a form witness is one determined by π, and the surrogate destroys exactly the fibre. This is a descent question, and it makes the first proposition a corollary rather than a finding.

3.2 Three propositions

Proposition 1 (Degenerate null)**.** *If* W *factors through the marginal, the null distribution of* W *is* $\delta_{W(X)}$*. Hence* $\overline{W} = W(X)$*,* $\sigma_W = 0$*, and* $z(W) = 0/0$ *is undefined.*

Proof. $W(gX) = W(X)$ for all $g \in G$, so the pushforward of any measure on G under $g \mapsto W(gX)$ is the point mass at $W(X)$. □

Corollary 1. *A reported small* z *for a factoring witness is not a small effect. Evaluated in floating point,* $0/0$ *returns the residue of summation order. Measured on the lattice of §4, where the mean of* $\operatorname{Re} U$ *over links is a symmetric function of the link multiset:* $\sigma_W = 6.5 \times 10^{-16}$ *and the computed* z *is* 1.4*.*

Proposition 2 (Reachable ceiling)**.** *Let* W *take values in* $[a, b]$ *with null mean* μ *and* $\sigma_W > 0$*. For every object with that null,*

$$z(W) \;\le\; C(W) \;:=\; \frac{\max(b - \mu,\ \mu - a)}{\sigma_W}.$$

If $C(W) < \tau$ *for threshold* τ*, no object attains* $z \ge \tau$*.*

Proof. $|W(X') - \mu| \le \max(b - \mu, \mu - a)$ since $W(X') \in [a, b]$. □

A low z from such a witness is a statement about resolution; calling it *form* attributes to the object what belongs to the test. Measured: a Wilson loop trace on a 6×6 lattice, range $[-1, 1]$, has $\sigma_W \in [0.11, 0.45]$, giving $C < 5$ and in places $C < 2$ against $\tau = 3$.

Proposition 3 (Non-factoring is not sufficient)**.** *Non-factoring is necessary for* $\sigma_W > 0$ *but does not imply* $W(X)$ *is separated from the null: there are non-factoring* W *and objects* X *with* $z(W)$ *arbitrarily small.*

Proof. Let $W(x) = \#\{i : x_{i+1} > x_i\}$, which does not factor. For X a discretised sine period of length n the sequence ascends on half its period, so $W(X) \approx n/2$, while the null mean is $(n-1)/2$. At $n = 60$: $W(X) \in \{29, 30\}$, null mean 29.5, $z = 0.34$. The same W on a monotone sequence gives $W = n - 1$ and large z. □

Hence *form* is a property of the pair (W, X), never of W. The practical consequence is a reading of [2]'s own results: its distributional witnesses are form *on the zeros*, which is what was measured, and are serviceable statistics on objects whose marginal does not determine them.

Remark 1. None of the three is novel as statistics: a permutation test needs a statistic that varies under the null, and any test needs power. The single error behind all three is conflating "does not factor" with "detects".

3.3 The decision procedure

Listing 1: The four-tier verdict (abridged from `src/shuffle-knife.ts`). `floor` is scaled to the witness magnitude; `range` is present only for bounded witnesses.

```
const floor = 1e-12 * Math.max(1, Math.abs(real));
const worst = Math.max(...vals.map((x) => Math.abs(x - real)));

// (1) structural: every surrogate IS the object, so there is no statistic
if (worst <= floor) return { tier: "FORM_EXACT", z: null, ceiling: null };

// every surrogate agrees with every other but not with the object
if (sd <= floor) return { tier: "FUNCTION", z: Infinity, ceiling: null };

const ceiling = w.range ? Math.max(w.range[1] - mean, mean - w.range[0]) / sd : null;
const z = Math.abs(real - mean) / sd;

// (2) the witness could not have reached the bar, so its z is not evidence
if (ceiling !== null && ceiling < threshold) return { tier: "NO_POWER", z, ceiling };

// (3) a verdict about THIS pair, not about the witness
return { tier: z >= threshold ? "FUNCTION" : "FORM", z, ceiling };
```

A separate check catches a different redundancy: witnesses pairwise affine across three or more objects are one witness twice, and counting both double-counts.

3.4 Calibration is a precondition

The caller supplies the surrogate construction, because only the caller knows what the marginal is; this is the one irreducibly domain-specific choice. If the construction fails to preserve the marginal it is randomisation, and then every witness reads FUNCTION for a reason unrelated to the object. *No internal check detects this and nothing about the output looks wrong.* Measured: replacing the permutation with fresh random values convicts the sample variance — a provable invariant — as FUNCTION. The only detector is the requirement that a known invariant return FORM_EXACT, which is why calibration is a precondition rather than good practice.

The implemented protocol: known invariants must read FORM_EXACT with z withheld; known non-factoring witnesses on a structured object must read FUNCTION; a decoy independent of the object must not read FUNCTION; a deliberately marginal-breaking surrogate must convict an invariant; fewer than two surrogates must refuse to run.

4 The lattice instrument

A $W \times W$ periodic lattice carries $U \in SU(2)$ on each link as a unit quaternion [5]. The marginal is the multiset of link rotors; the surrogate permutes which site each link occupies. Spatial adjacency is what gauge structure consists of — a plaquette being the ordered product around a face — so adjacency is the structure the surrogate must destroy.

4.1 A three-character action

Let $w = \frac{1}{2}\operatorname{tr} P$ for a plaquette P. The $SU(2)$ characters $\chi_{1/2} = 2w$, $\chi_1 = 4w^2 - 1$ and $\chi_{3/2} = 8w^3 - 4w$ give, normalised to unity at the identity, a per-face potential

$$g(w) \;=\; \underbrace{\beta(1-w)}_{\text{fundamental}} \;-\; \underbrace{\alpha(1-w^2)}_{\text{adjoint}} \;+\; \underbrace{\lambda\big(1-(2w^3-w)\big)}_{j=3/2}, \qquad V = \sum_{\text{faces}} g(w_P). \tag{1}$$

In the order parameter $m = 1 - w$ this is $c_1 m + c_2 m^2 + c_3 m^3$ with

$$c_1 = \beta - 2\alpha + 5\lambda, \qquad c_2 = \alpha - 6\lambda, \qquad c_3 = 2\lambda. \tag{2}$$

A single sign of λ therefore raises c_1, drives c_2 through zero and supplies $c_3 > 0$: the three things a first-order transition requires, with no shape fitted.

Every term is a different weight on one staple, which is why the force is one expression. For $V = \sum g(w_P)$,

$$F \;=\; \operatorname{vec}\Big(U \cdot \textstyle\sum_{\text{faces}} g'(w_P)\, s_{\text{face}}\Big), \qquad g'(w) = -\beta + 2\alpha w + \lambda(1 - 6w^2), \tag{3}$$

since $\sum \partial w_P / \partial U = -\operatorname{vec}(U s_{\text{face}})$. The sign of $g'(1) = -\beta + 2\alpha - 5\lambda$ reads the stability of the flat vacuum directly.

Listing 2: The force (abridged from `src/rotor-dynamics.ts`), with the pure-Wilson fast path retained so $\alpha = \lambda = 0$ is bit-identical to the prior organ.

```
export const faceWeight = (w, beta, alpha, lambda) =>
  -beta + 2 * alpha * w + lambda * (1 - 6 * w * w);

export function force(f, a, x, y, beta, alpha = 0, lambda = 0) {
  const u = getLink(f, a, x, y);
  if (alpha === 0 && lambda === 0) {            // every face carries one weight
    const v = vec(mul(u, staple(f, a, x, y)));
    return [-beta * v[0], -beta * v[1], -beta * v[2]];
  }
  const pair = staplePair(f, a, x, y);          // each face weighted at ITS own w
  const k0 = faceWeight(pair[0].w, beta, alpha, lambda);
  const k1 = faceWeight(pair[1].w, beta, alpha, lambda);
  const aw = [0,1,2,3].map((i) => k0 * pair[0].s[i] + k1 * pair[1].s[i]);
  return vec(mul(u, aw));
}
```

4.2 Validation

Equation (3) is checked against a central finite difference of V rather than against its derivation, at four (α, λ) pairs including a negative λ — a sign error cancelling for one sign would not survive both. Worst disagreement 2.4×10^{-9} at $\epsilon = 10^{-6}$. Dynamics are BAOAB Langevin; at $\gamma = 0, T = 0$ the integrator reproduces leapfrog to machine epsilon, and the position update being a group multiplication, $\|U\| = 1$ survives without renormalisation.

Four mutations of (1)–(3) were introduced and all are caught, with an informative split: mutants in g' kill seven tests by changing the dynamics, while mutants in V kill exactly one — the finite-difference check — because a potential/force mismatch is invisible to any sweep of the dynamics.

5 Criterion II: linear invisibility in parameter space

Proposition 4 (The quadratic spectrum sees one combination)**.** *Near the flat vacuum,* $m_P = 1 - w_P = \frac{1}{2}|v_P|^2 + O(|v|^4)$ *where* v_P *is the plaquette's algebra vector. Hence*

$$V \;=\; \frac{c_1}{2}\sum_P |v_P|^2 \;+\; O(|v|^4),$$

so the entire linearised fluctuation spectrum — every $\omega(k)$*, the whole small-amplitude response — is fixed by* $c_1 = \beta - 2\alpha + 5\lambda$ *alone, and* c_2*,* c_3 *first enter at* quartic *order in the field.*

Proof. m is already quadratic in the fluctuation, so $c_2 m^2$ and $c_3 m^3$ are quartic and sextic. □

Proposition 5 (Degeneracy)**.** *The family* $\{(\beta, \alpha, \lambda) : \beta - 2\alpha + 5\lambda = c_1\}$ *is two-dimensional. Any two of its members are indistinguishable to every measurement of quadratic response about the flat vacuum, and may differ in the number of minima of* g*, since that is governed by* $c_2^2 > 3c_1c_3$ *with* $c_2 < 0$*.*

Measured. Take $c_1 = \frac{1}{2}$ for both:

$$A = (\beta, \alpha, \lambda) = (0.5,\ 0,\ 0), \qquad B = (0.5,\ 1.0,\ 0.4).$$

A is pure Wilson with one minimum; B has $c_2 = -1.4$, $c_3 = 0.8$, hence $c_2^2 = 1.96 > 3c_1c_3 = 1.2$: two minima, the flat one metastable. Figure 1 measures the blindness as a *scaling exponent* rather than a tolerance.

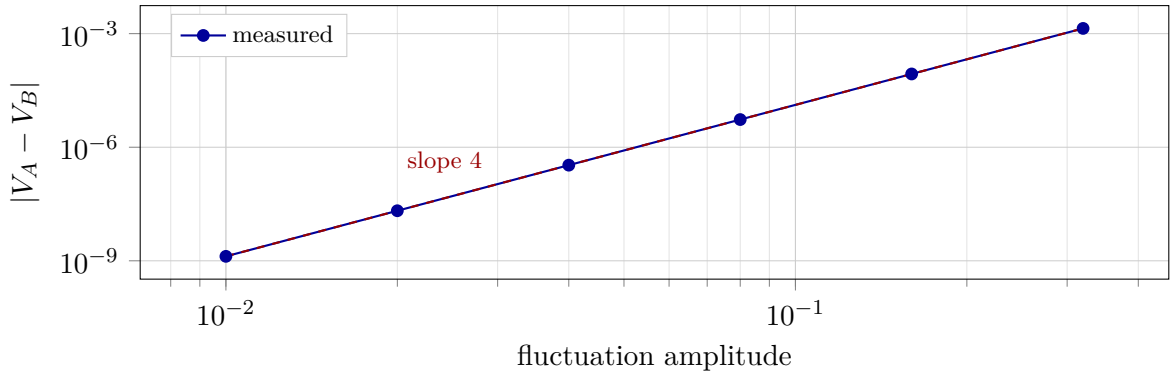


Figure 1: Two actions sharing c_1 disagree only at quartic order. Halving the amplitude divides the discrepancy by 16.0 at each of three successive halvings, and $|V_A - V_B|/\text{amp}^4$ is constant at 0.131. *Falsifier:* a quadratic disagreement would give a factor of 4. Breaking c_1 by 4% ($\beta = 0.5 \to 0.52$) moves the measured exponent from 16.0 to 4.0 — exactly quartic to quadratic — so the test detects the degeneracy's absence by kind, not by magnitude.

Relaxed from the flat vacuum both actions give 1.000000. Relaxed from the far basin, A returns to flat (0.999881) and B does not (0.263775). *Identical linear response, different vacuum count*, and the only measurement that separates them is one started in the other basin.

6 A first-order transition

The transition the cubic creates was pre-registered before measurement, including the metric and its kill condition, because the obvious metric is the wrong one: the largest step in a swept order parameter cannot distinguish a discontinuity from a steeper continuous curve, being a function of the sweep resolution. At $\lambda = 0$ the same curve reads 0.098 on a grid of 0.1 and 0.167 on a coarser one.

The discriminant is *hysteresis*. A damped relaxation sits in whichever minimum it started nearest until that minimum stops *existing* — the spinodal — not until it stops being global, so two minima make the up- and down-sweeps disagree and one minimum cannot, at any steepness.

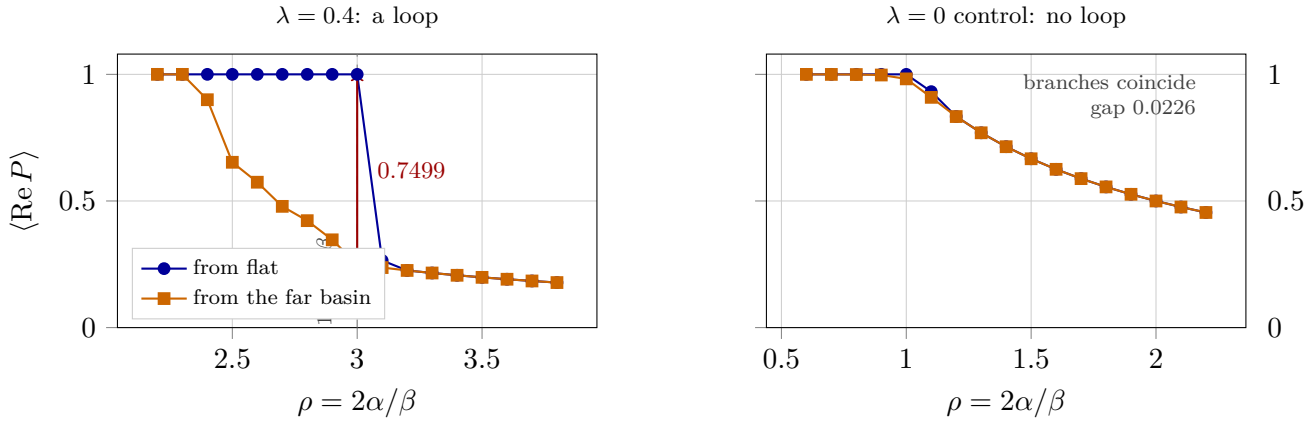


Figure 2: **Left:** hysteresis at $\beta = 1$, $\lambda = 0.4$, 6×6, 4000 damped steps. The branches differ by 0.7499 at $\rho = 3.00$ and the loop is about 0.5 wide in ρ. The flat vacuum survives to the predicted spinodal $\rho = 1 + 5\lambda/\beta = 3.00$, where $g'(1) = 0$ identically, and is gone by 3.10. **Right:** the same relaxation and seeds with the cubic switched off; the branches coincide, gap 0.0226, below the pre-registered void threshold of 0.05. One minimum cannot produce a loop at any steepness, so this control — and not the size of the jump — is the evidence. *Falsifier:* a gap below 0.05 on the left would retire the cubic as insufficient; a gap above it on the right would void the measurement.

Coexistence, and a scalar that lies. Between the branches the order parameter reads 0.263775. That value is attained by *no face*.

Reported as failed. The pre-registered convergence check did *not* pass. Doubling the relaxation at $\rho = 3.1$ moved the order parameter by 0.028, against the 0.02 allowed, drifting toward the $\rho = 3.2$ value — critical slowing down just above a spinodal. The jump's *magnitude* at that ρ is therefore unconverged and the suite asserts a floor rather than a value. The hysteresis result does not rest on it: both endpoints of the gap are converged.

What had not converged (added 23 October 2026). A finite-size-scaling run over widths 6 to 32 and three seeds found it was the step count, not the lattice: from 8000 steps every width and seed reads 0.237054, the smaller root of the per-face force $6\lambda w^2 - 2\alpha w + (\beta - \lambda) = 0$ at $\alpha = \rho\beta/2$, $\rho = 3.1$. The converged jump is therefore the relation $1 - w^*(3.1) = 0.762946$, measured against the $\rho = 2.9$ side where flat is exact, and the suite now asserts it to five decimals. The pre-registered verdict stands: at 4000 steps the reading was unconverged, and the 4000-step floor of 0.7 holds on the suite's lattice and seed but is missed at three of eighteen width and seed settings.

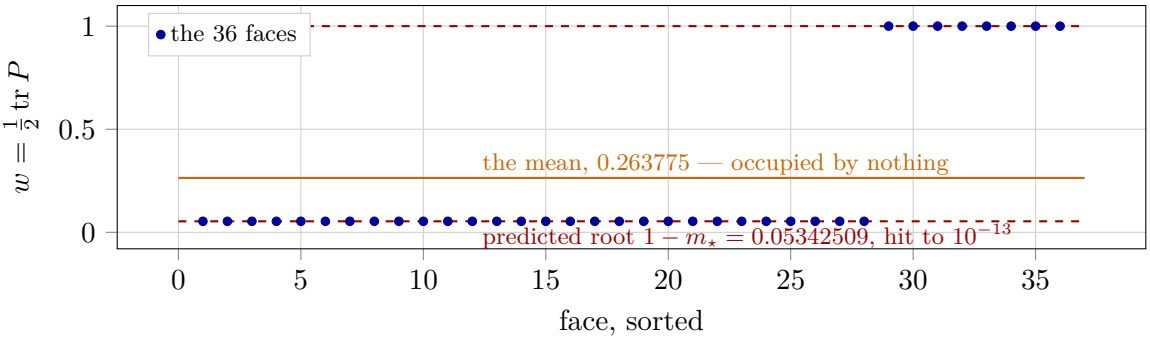


Figure 3: Two-phase coexistence with a pinned domain wall, at $\beta = 0.5$, $\alpha = 1.0$, $\lambda = 0.4$. 28 faces sit at the cubic's far root $1 - m_\star$ with $m_\star = (7\lambda - \beta)/(6\lambda)$, attained to 10^{-13}, and 8 at exactly 1; nothing lies between. The mean is the exact count-weighted average and is occupied by no member — read as a scalar it reports a partial collapse, which is false. Coexistence is the defining signature of a first-order transition and was not what the sweep was looking for. 6000, 24000 and 96000 steps give the same six decimals: at zero temperature a wall between barrier-separated minima has no downhill direction.

7 The battery, measured

Criterion I applied to the lattice, $K = 40$ surrogates, two configurations.

Witness	smooth configuration	two-phase	note
mean $\mathrm{Re}\,U$ (links)	FORM_EXACT	FORM_EXACT	factors; also gauge-dependent
max $\lvert\lVert U\rVert - 1\rvert$	FORM_EXACT	FORM_EXACT	factors
mean $\mathrm{Re}\,P$ (faces)	$z = 6.6$ FUNCTION	$z = 2.0$	
Wilson action at $\beta = 1$	$z = 6.6$	$z = 2.0$	affine to the row above
loop trace	$z = 1.1$	$z = 0.8$	NO_POWER
order sensitivity	$z = 0.6$	$z = 0.2$	NO_POWER

Six witnesses carry at most three readings. Two factor (Prop. 1). The Wilson action at $\beta = 1$ equals $W^2(1 - \langle \mathrm{Re}\,P\rangle)$, hence is affine to the face mean, and their z agree to nine decimals because they must. Two have $C(W) < \tau$ (Prop. 2). All four collapses are structural rather than statistical.

Remark 2. On the two-phase configuration the face mean sits $\approx 2\sigma$ from its own null, *below* the threshold: the coexistence structure of Fig. 3 is invisible to it and was visible only in the histogram. The link mean, meanwhile, has no consumer anywhere in the implementing codebase and the criterion supplies the reason — it is gauge-*dependent* (a gauge transformation moves it while the face mean holds to nine decimals) *and* factoring. Gauge-dependent together with factoring exhausts the ways a lattice witness can fail to inform.

8 Criterion III: localisation

8.1 The instrument, and its anchors

Riemann–Siegel Z with two remainder terms [4]; zeros by sign change and bisection; 2003 zeros on $t \in (20, 2520)$; unfolded by the Riemann–von Mangoldt term $\langle N(T)\rangle$. Anchored on values it was not tuned to:

Anchor	Published	Measured
θ at the first Gram point	17.8455995405	17.8455995408
first ten zeros	to 15 digits	worst deviation 2.5×10^{-3}
$N(50), N(100), N(300), N(1000)$	$10, 29, 138, 649$	**all exact**
$S(T)$ at those heights	$O(1)$, bounded	$+0.58, -0.00, +0.29, +0.38$

The two rows disagree in character and the distinction is load-bearing. Positions carry 2.5×10^{-3}, the remainder being asymptotic and $t \approx 14$ its weakest point. The *counts* are exact, and a reconstruction of the counting fluctuation depends on the count: a missed zero is invisible locally and would appear only as $S(T)$ drifting, which it does not. The instrument is fit for this use and unfit for anything needing a zero precisely.

The oscillating part of the explicit formula supplies the model,

$$S(T) \;\sim\; -\frac{1}{\pi} \sum_{p} \sum_{k \geq 1} \frac{1}{k}\, p^{-k/2} \sin\!\big(kT \log p\big), \tag{4}$$

whose frequencies $k \log p$ are *independent of* t; a fixed prime cutoff is therefore a fixed model at any height, which is what makes §8.3 meaningful. On a grid of 29456 points the correlation at primes <300 is 0.8362, against 0.840 reported in [3]; an amplitude-matched random-frequency control gives $+0.0035 \pm 0.0152$ against its 0.004 ± 0.019, placing the prime sum 55σ above the matched null against its $+54\sigma$.

8.2 The criterion

A truncated Fourier sum rings at the discontinuities of its target. If a residual is truncation, it is *localised* at those discontinuities; if it is missing structure, it is not. The test is therefore to mask a neighbourhood of each discontinuity and re-measure, with the kill condition fixed in advance: *no rise exceeding the cross-seed spread retires the truncation account.*

Listing 3: The mask. Distance to the nearest step is measured in *local* mean spacings, since the density of zeros grows with height.

```
const gap = ts.map((t, i) => {
  const d = nearestZeroDistance(t);            // |t - closest gamma|
  return d / meanSpacing(t);                   // 2*pi / log(t / 2*pi)
});
const away = ALL.filter((i) => gap[i] >= eps); // re-measure on the complement
const near = ALL.filter((i) => gap[i] <  eps); // and on what was removed
```

8.3 What the same mechanism explains

Reference [3] reports the reconstruction "flat in quality across a 50× range in the number of zeros" $(0.840, 0.848, 0.838, 0.840)$. Its table, however, raises the prime cutoff alongside M. At *fixed* cutoff the quality is not flat, and the decline is itself the step contamination: zeros are denser at greater height, so a larger share of any fixed grid falls within a ringing width of a step.

9 The battery on the zeros, and an amplification check

Criterion I on the zeros, $K = 40$, surrogate by permuting unfolded spacings:

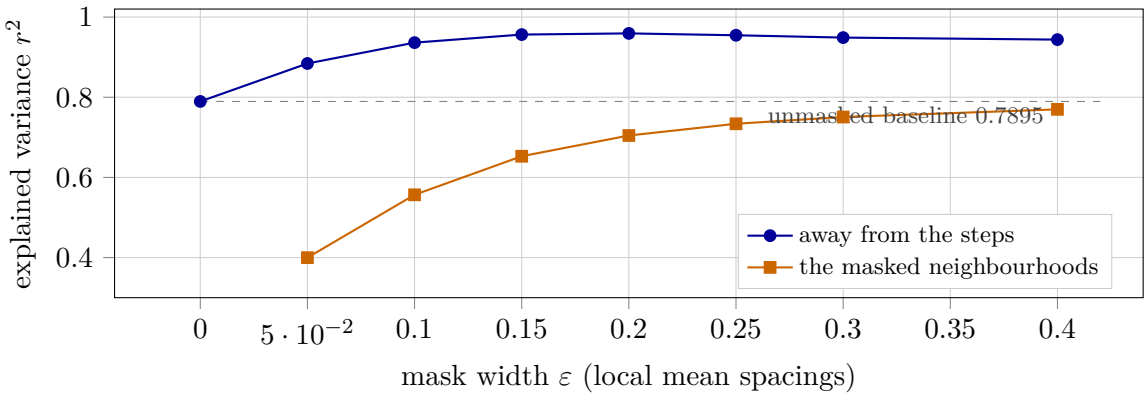


Figure 4: The residual of the explicit-formula reconstruction is localised at the staircase steps. Excluding points within $\varepsilon = 0.20$ of a zero raises r^2 from 0.7895 to 0.9593, while the excluded points read 0.3999 at $\varepsilon = 0.05$. Primes <3000, 29456 grid points. The rise eases past $\varepsilon \approx 0.2$ as the retained points concentrate mid-gap and the sample shrinks. *Falsifier:* no rise would locate structure the prime sum does not reach, retiring the truncation account and with it the completeness reading of [3].

Witness	σ_W	z	Tier
arithmetic, $\sum_{p\le 29}\langle\cos(\gamma \log p)\rangle^2$	3.51×10^{-4}	**499.8**	FUNCTION
lag-1 spacing correlation	2.16×10^{-2}	**17.6**	FUNCTION
spacing mean deviation	6.97×10^{-16}	withheld	FORM_EXACT
spacing variance	1.67×10^{-16}	withheld	FORM_EXACT
Wigner-shape L^2	exactly 0	withheld	FORM_EXACT

The split of [2] reproduces — the arithmetic witness carries the object-specific influence and the distributional witnesses carry none — and Prop. 1 changes the last three verdicts from *form at* 0.4–1.0σ to *factoring, significance withheld.*

Remark 3 (A correction to our own earlier figure). An earlier draft reported this agreement as 4%. That used the exponent rounded to 0.8, which moves the prediction from 450 to 481 and the discrepancy from 11% to 4%. The exact two-point exponent is 0.829 and the honest figure is 11%. The rounding flattered the comparison and is corrected here.

A second correction, 23 October 2026: the 11% was itself a single 40-surrogate draw at the eighty-third percentile of that estimate's spread. Converged, the measurement sits 7% below the prediction, and the prediction's own band, once the published points' sampling noise is propagated, contains both. Neither figure is a test of the exponent.

10 A removable artifact

Reference [2] attributes a 0.4–1.0σ residual in its factoring witnesses to its surrogate path: permute unfolded spacings, re-accumulate, then interpolate back onto the reference grid. The attribution is correct as to mechanism and localises further. Two routes from unfolded positions back to heights:

Route	spacing mean dev.	spacing variance	Wigner L^2
Analytic: bisect the monotone $\langle N\rangle$	6.2×10^{-16}	4.4×10^{-15}	exactly 0
Piecewise-linear between samples	1.9×10^{-16}	$\mathbf{1.03 \times 10^{-5}}$	$\mathbf{3.51 \times 10^{-4}}$

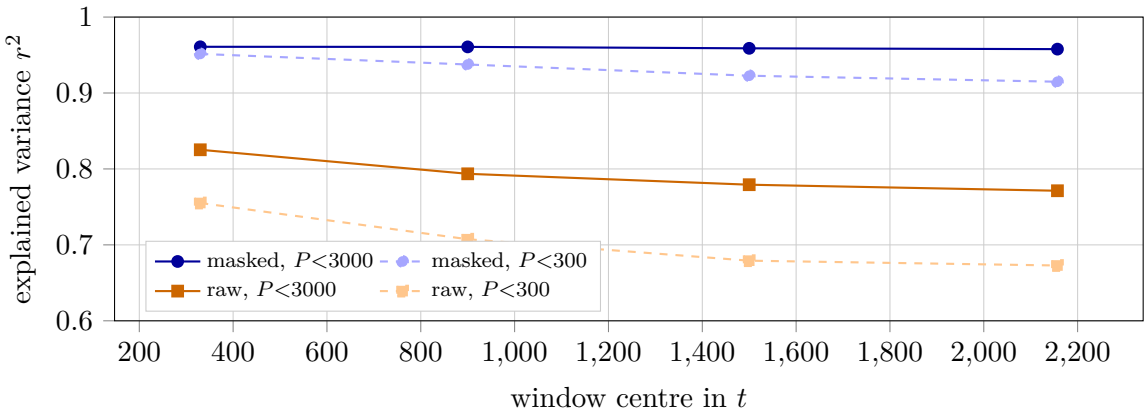


Figure 5: Masked, the reconstruction is scale-stable at adequate cutoff: 0.9609 → 0.9578 across a fourfold range in t, a decline of 0.003. Raw, it declines 0.7551 → 0.6728 at P<300. So the row-to-row constancy reported in [3] is held by growing the model with the data and is not evidence of stability at fixed complexity — while the stability it asserted is real, and is a property of the reconstruction away from the staircase.

The artifact requires interpolation between knots, not the round trip. Inverting the smooth counting function by bisection restores exact invariance, so the residual need not be explained — it can be removed. We do not compare the resulting z against the published 0.4–1.0: ours are 0.16 and 0.01, and z here is a ratio of one small quantity to another, dominated by grid, N and interpolation detail. The finding is σ_W moving from machine epsilon to 10^{-5}.

11 Discussion

The three criteria share a structure. Each identifies a *quotient* the measurement cannot see past — the marginal in §3, the quadratic order of response in §5, the location of a residual in §8 — and then asks whether the quantity of interest is determined by that quotient or lives in its fibre. Conflating "not determined by the quotient" with "detected" is the single error behind all of the omissions reported here.

The amendments cost something. Prop. 2 converts former *form* verdicts into *unresolved*, a loss of apparent information. The trade is favourable for an asymmetric reason: an unresolved verdict can be improved by a better witness or a larger object, whereas a false *form* verdict cannot be improved, because it does not know it is wrong. Prop. 1 replaces a number with a refusal, which is the harder amendment to accept, since a table with an omission looks weaker than a table of figures. It is the other way around.

Two of the measurements here are warnings about scalars rather than about instruments. The order parameter of Fig. 3 is exactly computed, carries no numerical flag, and denotes a state no member of its own distribution occupies; the apparent scale-degradation of Fig. 5 is a correctly computed decline that measures the grid's proximity to discontinuities rather than the model's reach. Neither failure announces itself. A number that overflows does not mislead — it fails loudly.

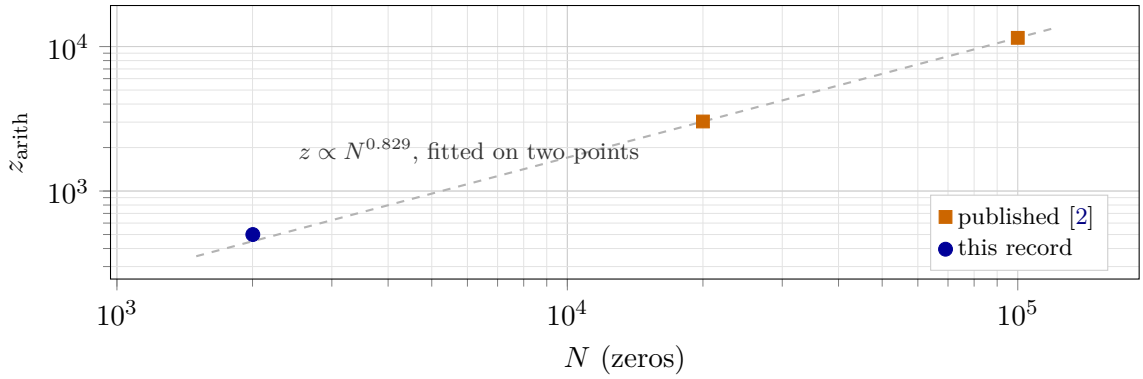


Figure 6: The function signal sharpens with N rather than averaging down, which is the argument that it resolves structure rather than an artifact. The two published points fix $z \propto N^{0.829}$; extrapolated a decade below the fitted range this predicts 450 at $N = 2003$, against 499.8 read at our seed. That reading is one draw of a 40-surrogate estimate: over 300 surrogate seeds it spans 331–580 (95%), and with 3000 surrogates it converges to 419.5 (95%: 406.5–433.2), 7% below the prediction. Treating the two published values as 60-surrogate estimates widens the prediction itself to 228–847 (95%). The growth with N is not in question; the exponent cannot be tested at this precision. The paired caveat in [2] holds in the stated direction: the *raw* value is larger at smaller N (0.1762 here against 0.0922 and 0.0636), so significance and magnitude move oppositely.

12 Scope and claim boundaries

What a function verdict establishes. That the witness responds to structure beyond the chosen quotient, for that object and that construction. Not that it is correct, meaningful, or evidence for any proposition about the object.

No claim about ζ or the Riemann hypothesis. Nothing here concerns the location of zeros off the critical line, the construction of a self-adjoint operator, or the completeness of the explicit formula as a theorem. Section 8 establishes where a *residual* lives, not that a series converges.

No claim about continuum $SU(2)$ or about confinement. Sections 4–6 concern a finite periodic lattice with a chosen three-character action at zero temperature. The transition is a property of that action on that lattice; no continuum limit is taken. A finite-size-scaling run at $\rho = 3.1$ over widths 6 to 32 (23 October 2026) finds the converged order parameter independent of width to six decimals; that measures the lattice dependence of this action, not a continuum limit.

Implementation dependence. Section 10 now rests on two independent implementations of each route (the second written from the first's interface only, with its own zeros, exact inverse and generator; 23 October 2026). Both find the residual present under piecewise-linear interpolation and at machine epsilon under exact inversion. Neither reproduces the published band's digits: over twenty seeds the second gives median z of 0.21 and 0.09. The mechanism is robust to implementation; the figures are not.

Unconverged where stated. The jump magnitude at $\rho = 3.1$ (§6) failed its pre-registered convergence check and is asserted as a floor only.

Where the methods stop. Criterion I requires a construction preserving the marginal while destroying correlation; where none exists the question is well posed and not answerable by this operator. Criterion III requires the instrument's failure mode to be *localised* in a known place; a residual spread uniformly is not separable this way.

13 Reproduction

Seed 20260423 throughout. Implementation: the `AISO-UI-for-Ai` tree (`frontend/src/shuffle-knife.ts`, `frontend/src/riemann-siegel.ts`, `frontend/src/rotor-field.ts`, `frontend/src/rotor-dynamics.ts`). The tree is public at https://github.com/tensorrent/AISO-UI-for-Ai; the numbers and figures in this document were produced on branch `claude/excel-memory-offloading-9ezwrv`. The pin is commit `55088af`: the four instrument files, the two zero-and-witness scripts and the lattice test are byte-identical from the generating commits (`086a0e3`, `64d9658`) through it, and the figure generator changed once after the generating run, at `55088af` itself, in the format of what it emits and not in any number (it regenerates the committed figure data byte-for-byte). Clone the repository and check it out at that commit,

```
git checkout 55088af96d4a3c282d3c8da44d7827ecbe904ddf
```

then `npm install` in `frontend/` and run the commands below from that directory.

Command	Produces
`npx vitest run tests/shuffle-knife.test.ts`	the calibration protocol, §3.4
`npx vitest run tests/riemann-siegel.test.ts`	the instrument anchors, §8
`npx vitest run tests/rotor-dynamics.test.ts`	§5–7, incl. the gradient check
`npx tsx scripts/riemann-witnesses.mts`	§9, both surrogate routes
`npx tsx scripts/riemann-positions.mts`	§8, §8.3
`npx tsx scripts/paper-figures.mts`	every figure in this paper

The last command emits `instrument_floors_data.tex`, which this document `\inputs`; no figure here is transcribed by hand. Scripts depending on an anchored instrument exit non-zero if an anchor fails.

A Dead ends and self-corrections

Recorded because two of the results above exist only as the residue of an error.

1. **The ascent count was predicted to read function.** Asserted in calibration; the assertion failed and the verdict was correct. Became Prop. 3.

2. **An analytic round trip was expected to reproduce the published band.** It does not — σ_W stays at machine epsilon — which showed "re-interpolation" too coarse a diagnosis and produced §10.

3. **The first destabilising term was link-based and had no threshold at all.** A term $-\alpha\sum(1-\operatorname{Re}U)$ is not gauge invariant, so it pushes on gauge directions, which have *zero* restoring force from β — that blindness being what gauge invariance is. Measured: order parameter at -1 by $\rho = 0.2$, collapse at $\alpha = 0.4$, no critical ratio anywhere. There is none when the attacked direction has no stiffness. The face-based term of (1) is the repair.

4. **A rounded exponent flattered a comparison** from 11% to 4% (Fig. 6); and the 11% was itself one 40-surrogate draw, 7% low once converged.

5. **Two implementations of an unrelated cyclicity figure disagreed by** 3.4σ, a spread as large as a discrepancy then being claimed against a published number. The claim was withdrawn until anchors settled it.

6. **A gauge-invariance control caught a malformed test.** A gauge transformation was supplied as a generator rather than a function of the vertex; `gaugeTransform` queries a vertex more than once, so each call returned a different rotor and the operation was not a gauge transformation. The control — $\langle \mathrm{Re}\, P \rangle$ must not move — detected it at $0.9961 \to 0.4202$.

References

[1] J. Theiler, S. Eubank, A. Longtin, B. Galdrikian, J. D. Farmer, *Testing for nonlinearity in time series: the method of surrogate data*, Physica D **58** (1–4), 77–94, 1992. doi:10.1016/0167-2789(92)90102-S.

[2] B. Wallace, *Spectral Rigidity and Shuffled Spacing Discriminant: Categorising Spectral Witnesses by What They Respond To* (FF06e), TensorRent, 2026; amendment of 24 September 2026.

[3] B. Wallace, *Density, Positions, Spacings: A Three-Layer Decomposition of the Riemann Zeros*, TensorRent, 2026; amendment of 24 September 2026.

[4] H. M. Edwards, *Riemann's Zeta Function*, Academic Press, 1974 (Riemann–Siegel formula, Ch. 7).

[5] K. G. Wilson, *Confinement of quarks*, Phys. Rev. D **10**, 2445–2459, 1974.

Contributions. The marginal-matched discriminant, the form/function vocabulary and the finding that a battery of agreeing witnesses may be one reading refracted are due to [2]; the three-layer decomposition and its completeness reading to [3]. Criterion I's three propositions and quotient recasting, the four-tier procedure, the three-character action and its validation, criteria II and III, both instruments, all six figures and the artifact localisation are this record's. Byline deliberately unset. Every figure is emitted by the measurement that produced it; no number here is quoted from memory or from a source not opened.